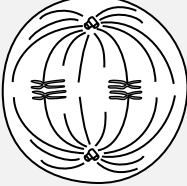
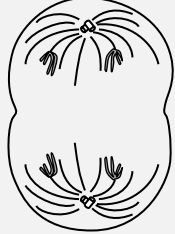


5(a)

| stage of meiosis | spindle fibres                                                                               | diagram                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| metaphase I      | attach to centromeres and arrange homologous pairs of chromosomes at the equator of the cell |                       |
| anaphase I       | (contract to) pull, centromeres / chromosomes, towards poles / to centrioles ;               |                       |
| prophase II ;    | re-form spindle in daughter cells                                                            |                      |
| telophase II     | disassemble                                                                                  | (forming) four daughter cells ;<br>two single chromosomes<br>inside a (re-forming) nuclear<br>envelope ; |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks    |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(b)     | <p>any <b>four</b> from:</p> <ol style="list-style-type: none"> <li>1 (meiosis / reduction division) produces gametes ;</li> <li>2 (two) gametes fuse / fertilisation occurs, to form a zygote ;</li> <li>3 zygote will have maternal and paternal chromosomes / AW ;</li> <li>4 gametes, are haploid / are <math>n</math> / have half the normal number (of chromosomes) ;</li> <li>5 so the zygote, is diploid / <math>2n</math> ;</li> <li>6 prevents, doubling of chromosome number / polyploidy / having too many chromosomes<br/><b>or</b><br/>allows chromosome number to remain constant ;</li> </ol> | <b>4</b> |

| Question | Answer | Marks |
|----------|--------|-------|
|----------|--------|-------|